1️⃣ What is Azure Virtual Network (VNet)?

Azure Virtual Network (VNet) is a logically isolated network in Azure that enables secure communication between:

- Azure resources (VMs, App Services, etc.)
- Internet
- On-premises networks
- Other VNets

Think of it like your **private data center network** inside Azure.

Main Components:

Component	Description
Address Space	CIDR block assigned to VNet (e.g., 10.0.0.0/16)
Subnets	Logical segmentation of VNet
NIC (Network Interface)	Connects VM to subnet
Public IP	Allows internet communication
NSG	Controls inbound/outbound traffic
Route Table (UDR)	Controls traffic routing
Gateway Subnet	Used for VPN/ExpressRoute

2️⃣ Subnets – *Network Segmentation*

What is Subnet?

A **logical division** inside a VNet.

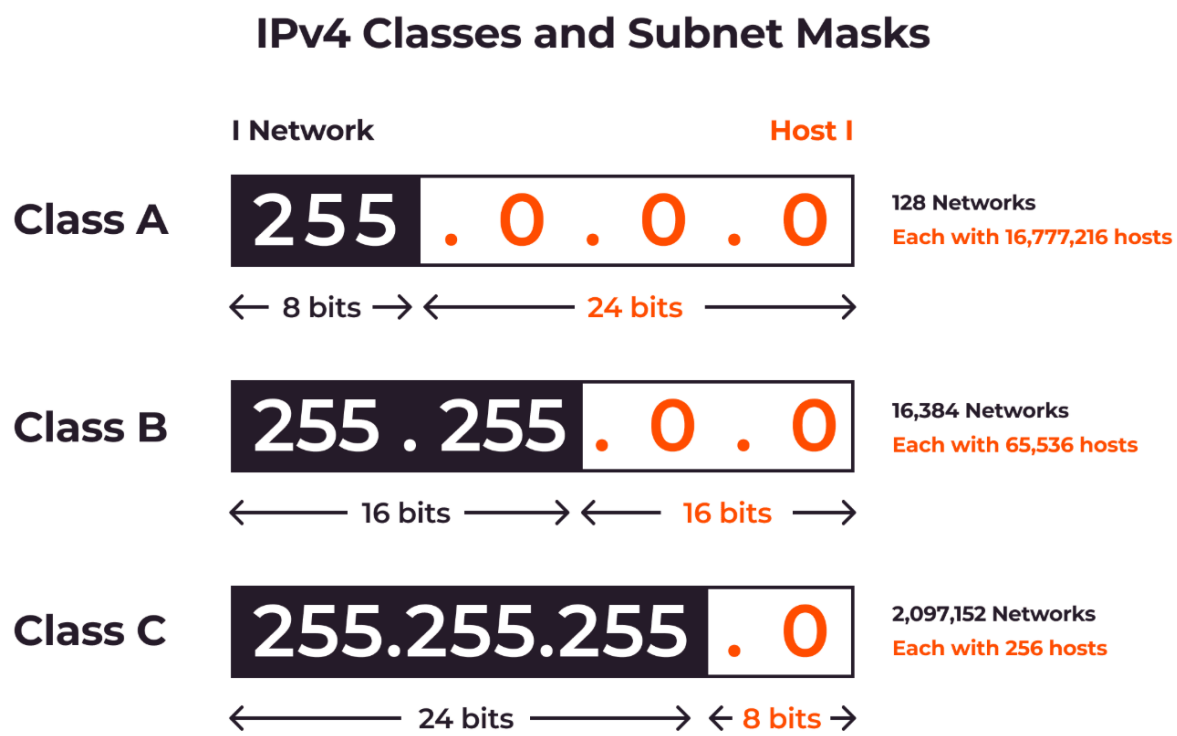
Why use subnets?

- Security isolation
- Apply **NSG**, **Route Table**, **Firewall**
- Separate tiers (Web / App / DB)

Example

```makefile
VNet: 10.0.0.0/16

Subnet-Web: 10.0.1.0/24
Subnet-App: 10.0.2.0/24
Subnet-DB:  10.0.3.0/24
```



Subnet Types

Subnet Type	Purpose
Default Subnet	General resources
GatewaySubnet	VPN/ExpressRoute
AzureBastionSubnet	For Bastion
AzureFirewallSubnet	For Azure Firewall
AppGatewaySubnet	For Application Gateway

3️⃣ IP Addressing in Azure

Types of IPs

Type	Use
Private IP	Internal communication
Public IP	Internet access

Allocation

- **Static** – Fixed IP (recommended for servers)
- **Dynamic** – Changes on reboot

Real-World Tip

- VM in production → **Static private IP**
- Load balancer → **Static public IP**

Steps to Create VNet `10.0.0.0/16` with Two Subnets (AZ-104 Lab)

We will create:

- **VNet:** `10.0.0.0/16`
- **Subnet1:** `10.0.1.0/24`
- **Subnet2:** `10.0.2.0/24`

Step 1️⃣: Login to Azure Portal

Go to:

👉 https://portal.azure.com ↗

Search for **Virtual networks** in the top search bar.

Step 2️⃣: Create Virtual Network

🟢 Basics Tab

Fill:

- **Subscription:** Select your subscription
- **Resource Group:** Create new (Example: `RG-Network-Lab`)
- **Name:** `VNet-Lab`
- **Region:** Choose same region for all resources (Important for exam)

Click **Next**

IP Addresses Tab

Under **IPv4 address space**

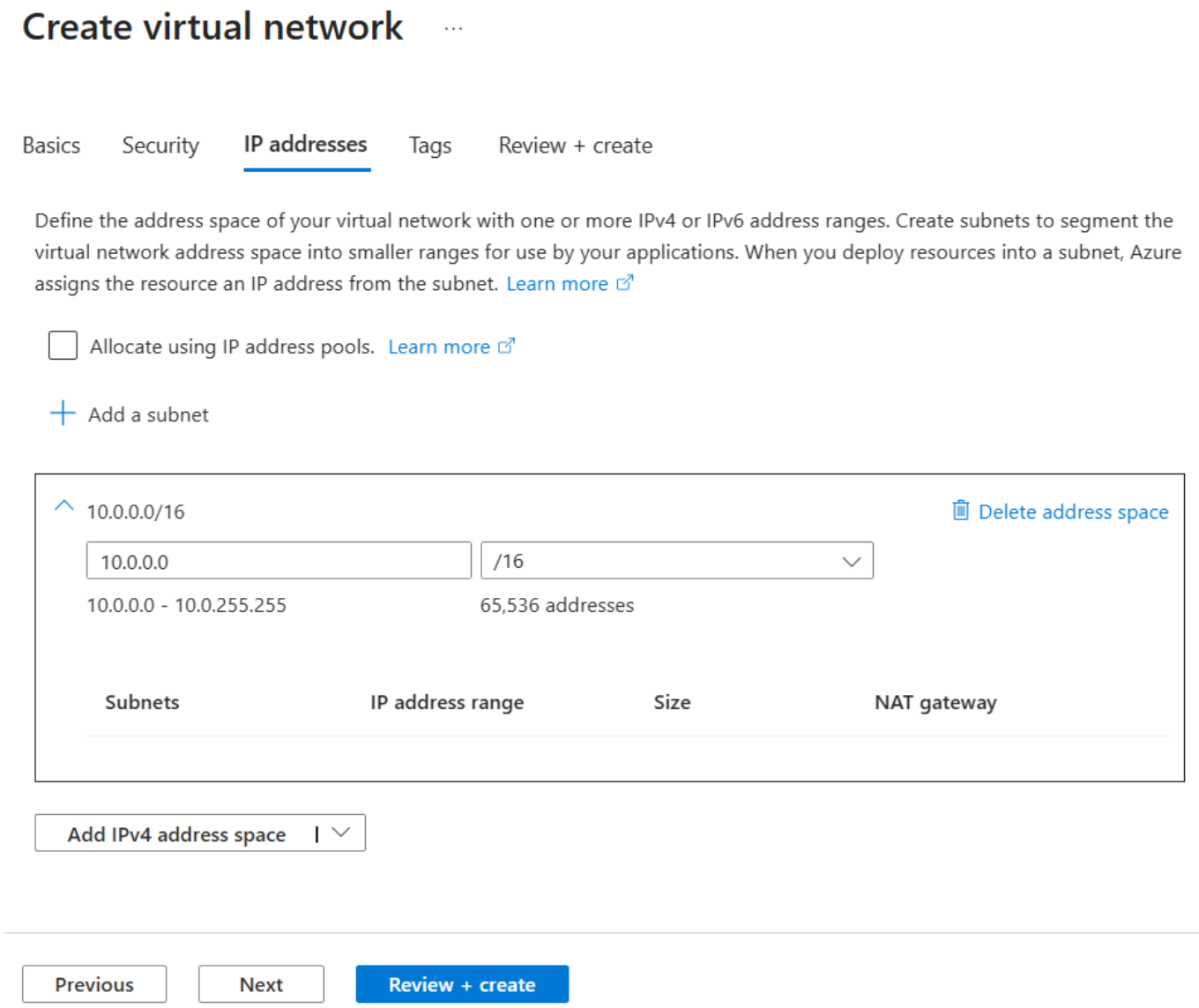
- Address space:

```
Code

10.0.0.0/16
```

Remove default subnet if present.

Now click + **Add subnet**



Step 3️⃣: Create Subnet1

- Subnet name: `Subnet1`
- Subnet address range:

```
Code

10.0.1.0/24
```

Click **Add**

Step 4️⃣: Create Subnet2

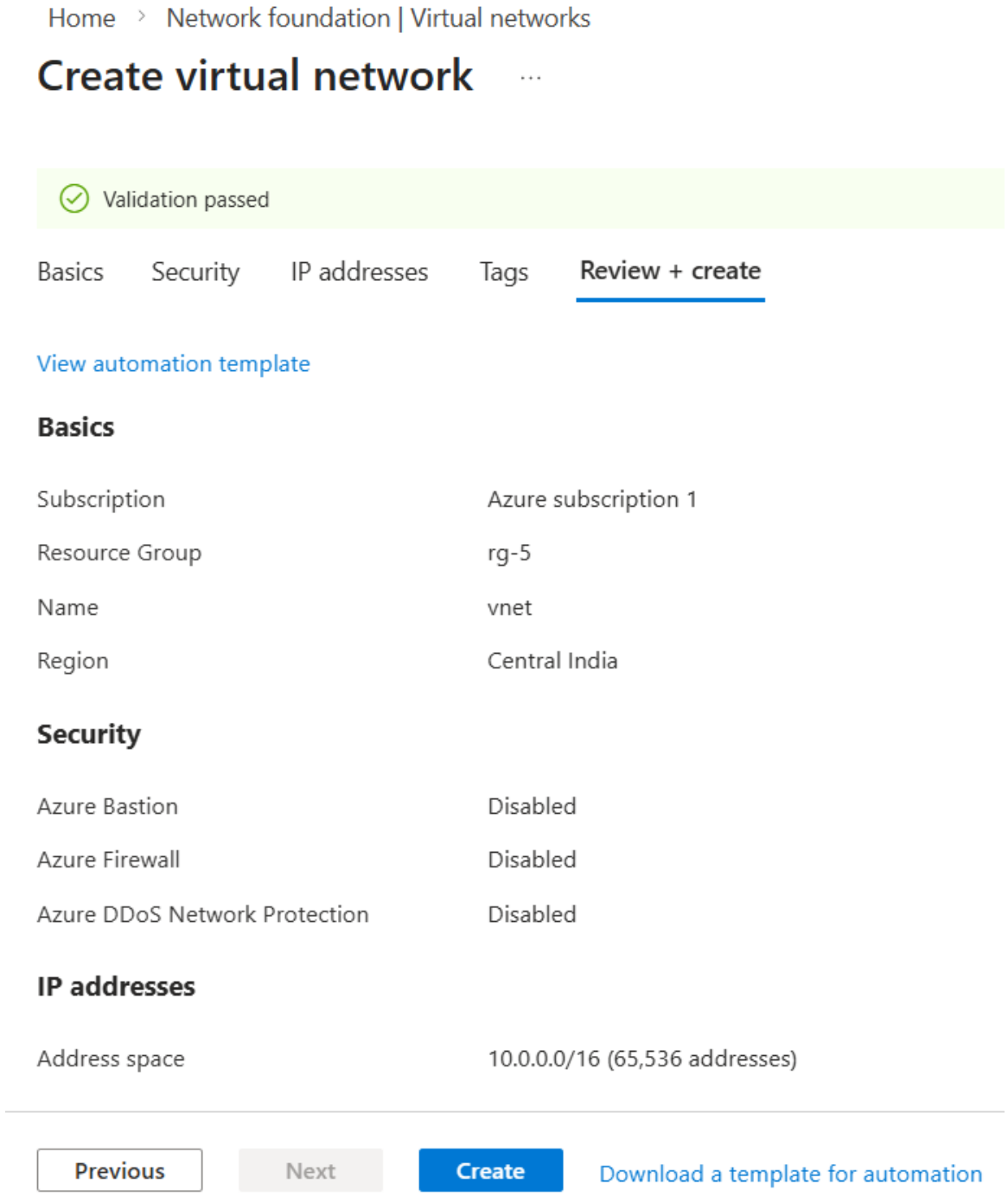
Click + **Add subnet**

- **Subnet name:** `Subnet2`
- **Subnet address range:**

```
10.0.2.0/24
```

Click **Add**

Now click **Review + Create**

Then click **Create**

Deployment will take 1–2 minutes.